

HISTORY OF COMPUTERS

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ABACUS

- Around 1000 years before the birth of Jesus Christ the Mesopotamians quite unknowingly laid the foundation of computer era. They made a device of stones sand and pebbles, called Abacus. It was mainly used for counting.

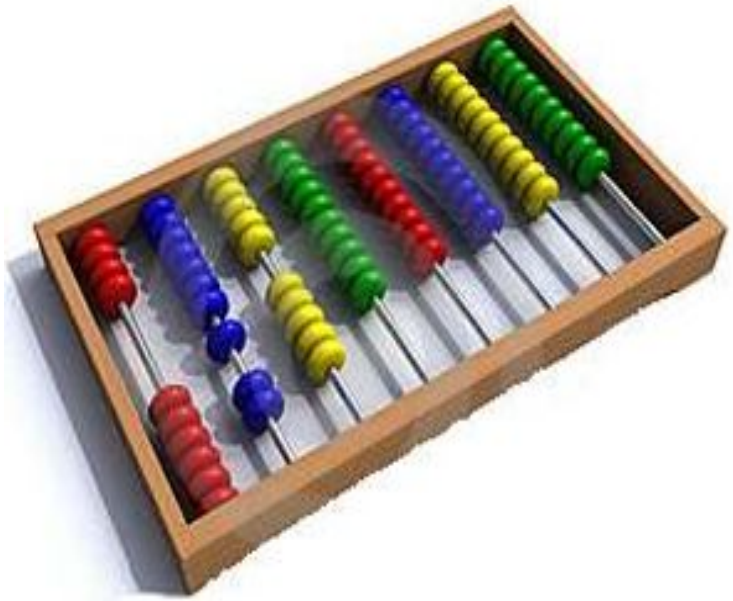
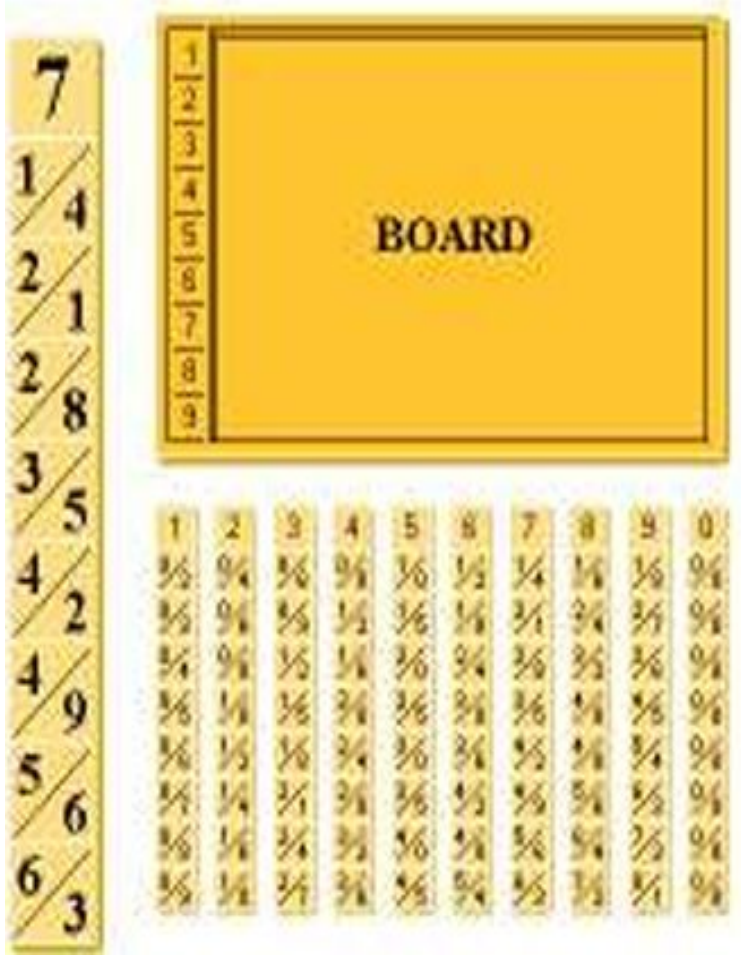


Figure 1.1

NAPIER'S BONES

- John Napier was a Scottish mathematician who lived from 1550 to 1670. He worked for more than 20 years to develop the theory of tables, which became quite famous, with the name Logarithmic Tables. Towards the end of his life, John Napier devised set of rods, called Bones, because they were constituted from bones. Digits were printed on them. Napier's rods essentially transformed the process of complex multiplication and division into much simpler tasks of addition and subtraction.



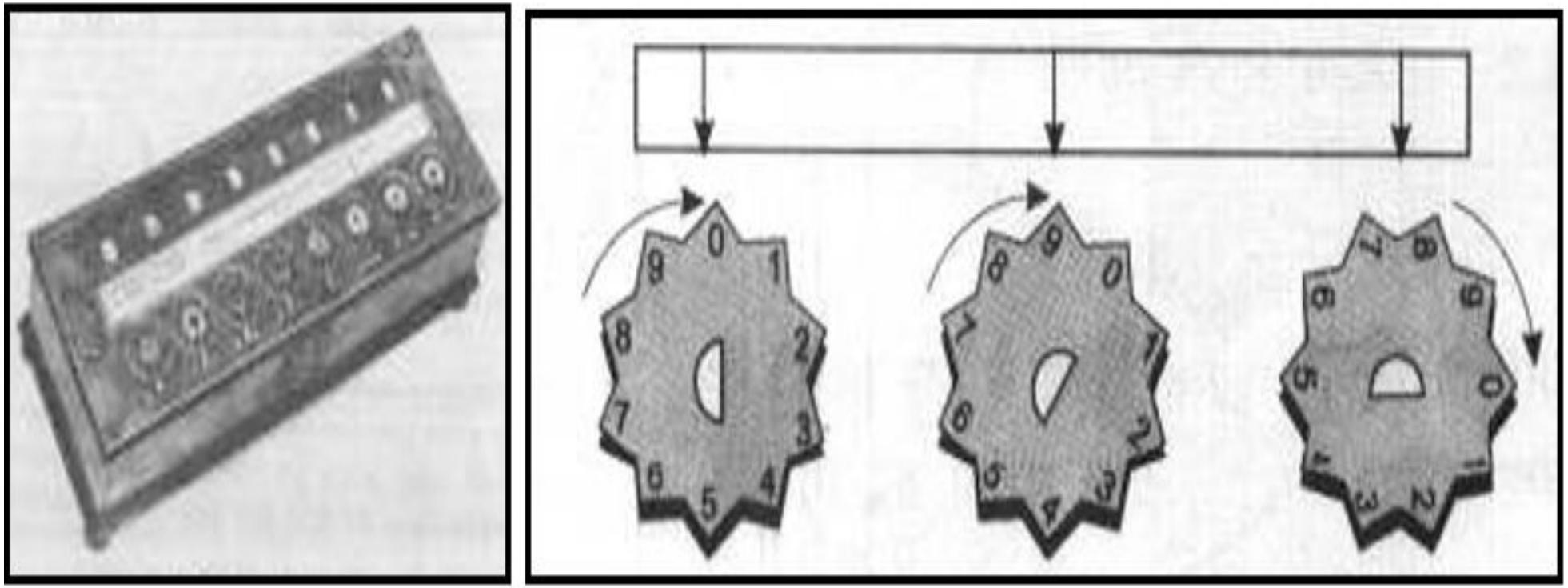
SLIDE RULE

- Slide rule was once-upon-a-time favorite mechanical device that was quite often used for performing complex and complicated calculations at very fast speed. Using slide rule mathematical, statistical and engineering calculations are performed with ease. For doing all these calculations, it makes use of logarithmic tables that were invented by John Napier in 1941. The slide rule consists of a pair of finely divided fixed scales. Most often, outer pair of scales is fixed and inner scale is movable. A sliding window called cursor is put around the scales.

PASCAL'S CALCULATOR

- Blaise Pascal was a French mathematician, physist and theologian.
- He was the son of tax collector. His father had to do lot of additions and subtractions, in which he quite often made errors.
- At the age of 18 in the year 1840 Pascal started developing a mechanical device, using which arithmetic operations like addition and subtractions could be done very easily. The first operating model was introduced in 1942.
- This machine was called Arithmetic Machine. Later Pascal created 50 more machines, over next 10 years. Later this machine was called, Pascal's Adding Machine.

PASCAL'S CALCULATOR

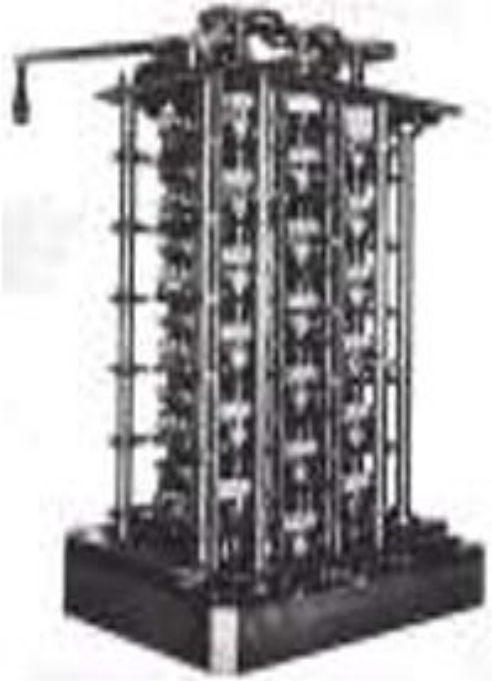


LEIBNIZ'S CALCULATOR

- A German mathematician, whose name was Gottfried Leibniz, improved the adding machine, developed by Pascal. This new machine, which he constructed in year 1671, was able to multiply and divide as well. The concept of multiplication was to repeatedly add the same number to itself. Instead of using wheels in his machine, he used stepped cylinders, each with nine teeth of varying lengths.

BABBAGE'S DIFFERENCE ENGINE

- In year 1822, Charles Babbage, a professor of mathematics developed a machine called Difference Engine. This machine was able to calculate logarithms up to high precision degree. It had capability to calculate different functional values and could also calculate polynomials by finite difference technique.

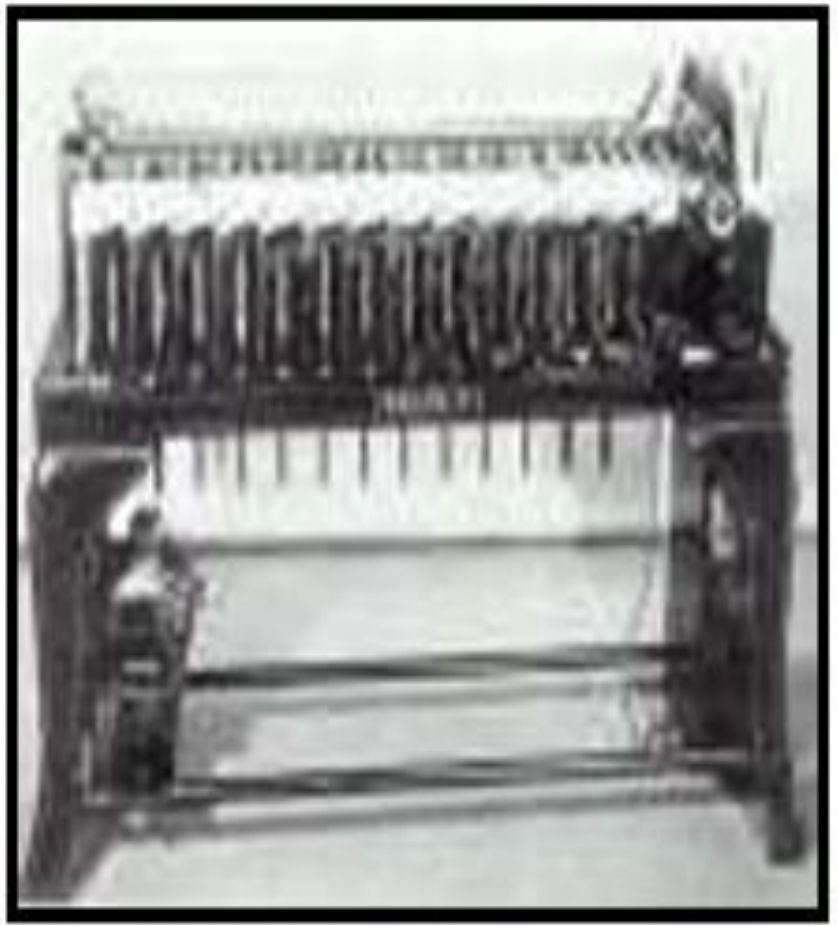


BABBAGE'S ANALYTICAL ENGINE

- Charles Babbage started working on a device named Analytical Engine in year 1833.
- This Analytical Engine was in real terms the ancestor of modern computer.
- The Analytical Engine was able to perform all four arithmetic operations.
- It was conceptualized around processor, storage area, memory and input/output devices.

HOLLERITH'S MACHINE

- The dream of Charles Babbage was fabricated by an American, named Herman Hollerith (1869-1926).
- In year 1887 he fabricated the first electromechanical punched card tabulator that used punched cards for input, output and instructions.
- This machine was used by American department of census to compile their 1880 census data and was able to complete compilation in 3 years, which earlier used to take around 10 years.



MARK-1

- Prof Howard Aiken (1900-1973) of U.S.A constructed an electromechanical computer named Mark-1 in year 1943. Mark-1 had capability to multiply two 10-digit numbers in 5 seconds, which was a record at that time. Mark-1 was the first machine, which could perform according to pre programmed instructions automatically without any manual interference. This was the first operational general-purpose computer.



ENIAC

- The defense department of USA was supposed to design advanced weapons, missiles and aircrafts. For this, they had to do lot of complex calculations.
- This process was not only time consuming but was error prone also. To do this work efficiently, in 1946, scientists of Pennsylvania University designed a new computer.
- This computer was 20 feet long and 40 feet high. It consumed as much electric power as was sufficient to light a small city. But this machine was much faster than any earlier machine. It was capable of doing more than 5000 additions, subtractions or 350 multiplications and divisions in a second. This machine was called **ENIAC. (Electronic Numerical Integrator and Computer.**)

ENIAC



EDVAC

- EDVAC is acronym for Electronic Discreet Variable Automatic Computer.
- It was designed and created in 1950 in Moore school of America. It was capable of taking instructions from external resources.
- Data and instructions, both could be provided to it, using punched cards. The problem of hardwired instructions, which was there in ENIAC, was solved in this computer. Its performance was much better than earlier computers.

MODERN COMPUTER

- In the year 1981, IBM developed a modern computer. It was small in size but very powerful. Its functioning was very simple. This computer was called Personal Computer or PC. It became popular very soon. After 1981 lot of improvements were made in PC. These improvements made PC very powerful and easy to operate. Personal Computer's functioning has now become so simple that even a child can operate it.

